Rodrigo G. Freundt

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Summary

- Research-oriented engineer and Ph.D. candidate with +10 years of technical experience in astronomical instrumentation. Excellent technical writing, project management and mentoring skills. Extensive experience working in large collaboration with diverse teams of scientists and engineers.

Education

Cornelll University — Ithaca, NY
Ph.D. in Astronomy and Space Sciences. — 2021–Present

- Thesis: "The Epoch of Reionization Spectrometer (EoR-Spec): Development and Early Science." (in preparation). Expected completion: Spring 2027.

Pontifical Catholic University of Peru — Lima, Peru
B.Sc. in Electronics Engineering. — 2011–2016

- Thesis: "Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU".

Experience

CCAT Observatory — Ithaca, NY.
Graduate Research Assistant. Advisors: Prof. Gordon Stacey and Prof. Michael Niemack — 2021 –Present

- Developing state-of-the-art submillimeter instrumentation for the CCAT collaboration.
- Leading the design, fabrication and deployment of EoR-Spec, a direct-detection imaging spectrometer.
- In charge of the development of thermometry and cryogenic readout harnesses for the Prime-Cam receiver.

Atacama Cosmology Telescope — San Pedro de Atacama, Chile.
Site Engineer. Advisors: Prof. Suzanne Staggs and Prof. Mark Devlin. — Oct 2019 –Jul 2021

- Ensured the continuous operation of complex astronomical instrumentation at extreme altitude (17,000 ft). In charge of regular maintenance, troubleshooting and overhaul of electrical, cryogenic and mechanical systems.
- Performed calibration procedures, such as the alignment of the receiver with the primary and secondary reflectors through high-precision photogrammetry ($\sim 10\mu$m). Carried out seasonal passband characterization via Fourier Transform Spectroscopy.
- Helped in the deployment of the last detector array upgrade (27/39 GHz) for the ACTPol receiver.

Management Solutions — Lima, Peru / Madrid, Spain.
Assistant Business Consultant — Sep 2018 –Jul 2019

- Worked in big data and bank stress test for a major local bank in Lima, Peru. Software development and data visualization with interactive dashboards for a leading financial institution in Madrid, Spain.

Institute for Radioastronomy — Lima, Peru.
Research Engineer (Intern until Apr 2017) Advisor: Prof. Jorge Heraud — Mar 2015 –Aug 2018

- Worked in the construction of an 8-meter radio telescope (RT8) intended for teaching and outreach. Designed hardware and software for the telescope's electromechanical pointing system.
- Developed custom firmware for the backend electronics of the telescope.
- Assisted in the characterization of a 1.42 GHz feed antenna using software-defined radio technology.

Skills

- **Programming:** Python, C/C++, CUDA, LabVIEW, microcontroller firmware.
- **CAD/Simulation** Solidworks, KiCAD, Ansys Zemax Optics Studio.
- **Tools:** LaTeX, Git, Docker, Linux.

Awards

- Beca de Estímulo Académico Solidario (BEAS) 2013–2016
  Fully-funded undergraduate scholarship for outstanding academic performance and financial need.
- URSI Young Scientist Award Aug 2017
  Merit-based registration fee waiver and travel grant to attend URSI 2017, Montreal, QC.
- Summer School of Astronomical Instrumentation, Dunlap Institute, UofT Jul 2019
  Merit-based tuition waiver and travel grant. Toronto, ON.
- CASPER Workshop, Harvard & Smithsonian Center for Astrophysics (CfA) Aug 2019
  Merit-based fee waiver and travel grant (invited, could not attend). Cambridge, MA.

Teaching

- **Teaching Assistant** at Cornell University Fall 2022, 2024, 2025, 2026
  Multiwavelenght astronomical techniques (ASTRO 4410)
- **Teaching Assistant** at Cornell University Spring 2023
  Our Solar System (ASTRO 1102)
- **Teaching Assistant** at PUCP Aug 2017 - Dec 2017
  Space Science and Engineering (ING 306)

First-Author Publications

[1] B. Keller, **R. Freundt**, J. R. Burgoyne, S. Chapman, S. Choi, C. J. Duell, C. Groppi, C. Humphreys, L. T. Lin, A. Middleton, M. D. Niemack, D. Patel, E. Vavagiakis, S. Walker, Y. Wang, and R. Xie, "Ccat: Flexible stripline circuits for large-format kinetic inductance detector (kid) array readout", *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 1–6, 2026.

[2] **R. Freundt**, Y. Li, D. Henke, J. Austermann, J. R. Burgoyne, S. Chapman, S. K. Choi, C. J. Duell, Z. Huber, M. Niemack, T. Nikola, L. Lin, D. A. Riechers, G. Stacey, A. K. Vaskuri, E. M. Vavagiakis, J. Wheeler, and B. Zou, "CCAT: A status update on the EoR-Spec instrument module for Prime-Cam", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 343–352.

[3] **R. Freundt** and J. A. Heraud, "Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU", in *2017 XXXIInd General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS)*, Aug. 2017, pp. 1–4.

Collaboration Publications

[4] Calabrese, E. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 constraints on extended cosmological models", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 063, Nov. 2025, ISSN: 1475-7516.

[5] Louis, T. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 power spectra, likelihoods and lcdm parameters", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 062, Nov. 2025, ISSN: 1475-7516.

[6] Middleton, A. et al. (including **R. G. Freundt**), "Ccat: Led mapping and characterization of the 280 ghz tin kid array", *IEEE Transactions on Applied Superconductivity*, vol. 35, no. 5, pp. 1–4, 2025.

[7] Naess, S. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 maps", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 061, Nov. 2025, ISSN: 1475-7516.

[8] Vaskuri, A. K. et al. (including **R. G. Freundt**), "280-GHz aluminum MKID arrays for the Fred Young Submillimeter Telescope", *Journal of Astronomical Telescopes, Instruments, and Systems*, vol. 11, 026005, p. 026 005, Apr. 2025.

[9] Coulton, W. et al. (including **R. G. Freundt**), "Atacama cosmology telescope: High-resolution component-separated maps across one third of the sky", *Physical Review D*, vol. 109, no. 6, Mar. 2024, ISSN: 2470-0029.

[10] Madhavacheril, M. S. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 gravitational lensing map and cosmological parameters", *The Astrophysical Journal*, vol. 962, no. 2, p. 113, Feb. 2024, ISSN: 1538-4357.

[11] Qu, F. J. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: A measurement of the dr6 cmb lensing power spectrum and its implications for structure growth", *The Astrophysical Journal*, vol. 962, no. 2, p. 112, Feb. 2024, ISSN: 1538-4357.

[12] B. Zou, J. R. Bond, V. Butler, **R. Freundt**, Z. B. Huber, M. D. Niemack, T. Nikola, G. J. Stacey, and E. M. Vavagiakis, "CCAT: Characterizing the Fabry-Pérot interferometer mirrors and mount for the epoch of reionization spectrometer", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 909–916.

[13] Choi, S. K. et al. (including **R. G. Freundt**), "Ccat-prime: Characterization of the first 280 ghz mkid array for prime-cam", *Journal of Low Temperature Physics*, vol. 209, no. 5-6, pp. 849–856, Aug. 2022, ISSN: 1573-7357.

[14] Collaboration, C.-P. et al. (including **R. G. Freundt**), "CCAT-prime Collaboration: Science Goals and Forecasts with Prime-Cam on the Fred Young Submillimeter Telescope", *The Astrophysical Journal Supplement Series*, vol. 264, no. 1, p. 7, Dec. 2022, et al. including R. G. Freundt, ISSN: 0067-0049.

[15] T. Nikola, S. K. Choi, C. J. Duell, R. G. Freundt, Z. B. Huber, Y. Li, K. Malavalli, M. Niemack, K. M. Rossi, G. J. Stacey, E. M. Vavagiakis, B. Zou, N. F. Cothard, J. Austermann, J. D. Wheeler, J. Gao, M. R. Vissers, J. Hubmayr, J. Beall, and J. Ullom, "CCAT-prime: The epoch reionization spectrometer for primce-cam on FYST", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI*, vol. 12190, SPIE, Aug. 31, 2022, pp. 247–262.

[16] B. Zou, S. K. Choi, N. F. Cothard, **R. Freundt**, Z. B. Huber, Y. Li, M. D. Niemack, T. Nikola, D. A. Riechers, K. M. Rossi, G. J. Stacey, and E. M. Vavagiakis, "CCAT-prime: The design and characterization of the silicon mirrors for the Fabry-Perot interferometer in the Epoch of reionization spectrometer", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI*, vol. 12190, SPIE, Aug. 31, 2022, pp. 982–994.